

Enter the following code in the Thonny Integrated Development Environment (IDE) and save it with a name and in the location of your choice:

```
counter = 0
while counter < 5:
    print(counter, end=" ")
    counter += 1
print("\nThe loop has ended.")
```

1. Set a breakpoint on the while statement (add line numbers first from **Tools, Options, Editor**) then double-click on the while line.

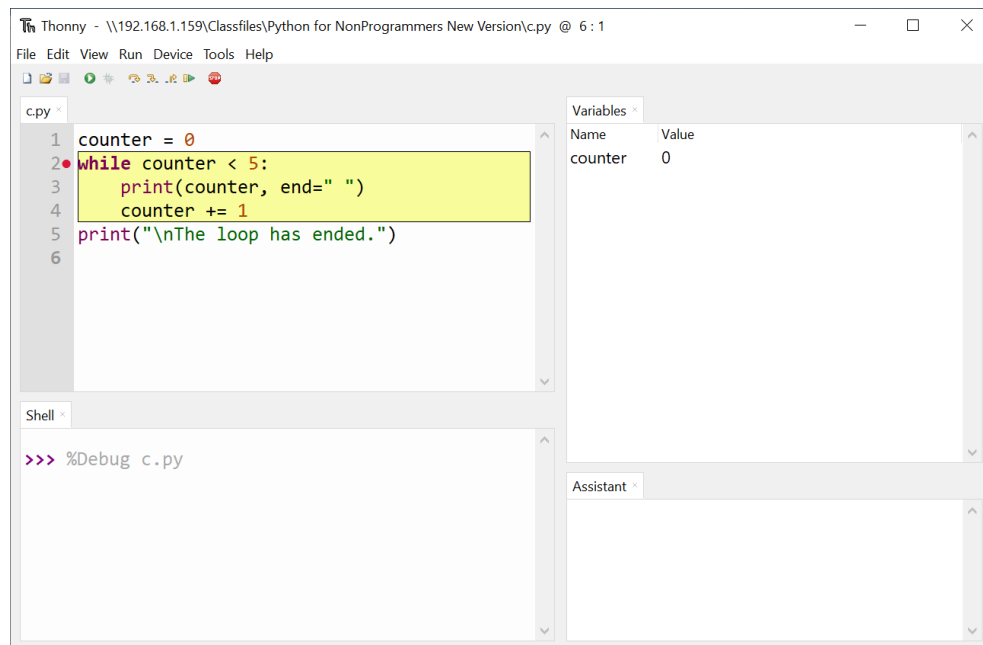
2. From the **View** menu add **Variables**.

3. Click on the bug icon  to the right of the green arrow to go to **Debug Mode**.

4. The counter variable will be created with a value of 0 and stop on the while statement without executing it.

5. You should also see the variable counter with a value of 0 added to the **Variables** window.

6. Notice that the entire while loop is highlighted in **yellow**.



7. Click once on the first downward arrow (step over or F6) to the right

of the bug icon  on the toolbar.

8. The while loop will execute 5 times printing 0 to 4 and stop at the next line, the print statement.

9. Press the same arrow  or F6 once more to execute the print statement.

10. Click on the **STOP** icon on the toolbar.

11. Run the program again by clicking on the bug icon.

12. Once again it will stop at the while line.

13. This time click on the second downward (step into or F7) facing arrow  step into.

14. This will allow you to execute each line within the while loop.

15. You will see the value of counter change for 0 to 4 in the **Variables** windows and also in the shell window.

16. Keep clicking the step into  or F7 until it exits the while loop.

17. Then use the step over or step into to execute the final print and end the program.